# Manage cluster peers

ONTAP 9.7 REST API reference

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# Manage cluster peers

## Cluster peers endpoint overview

### Overview

Cluster peering allows administrators of ONTAP systems to establish relationships between two or more independent clusters. When a relationship exists between two clusters, the clusters can exchange user data and configuration information, and coordinate operations. The /cluster/peers endpoint supports create, get, modify, and delete operations using GET, PATCH, POST and DELETE HTTP requests.

### Create a cluster peer

You can set up a new cluster peer relationship by issuing a POST request to /cluster/peers. Parameters in the POST body define the settings of the peering relationship. A successful POST request that succeeds in creating a peer returns HTTP status code "201", along with the details of the created peer, such as peer UUID, name, and authentication information. A failed POST request returns an HTTP error code along with a message indicating the reason for the error. This can include malformed requests and invalid operations.

#### Sample request

```
curl -X POST 'https://<mgmt-ip>/api/cluster/peers/' -d
'{"authentication":{"expiry_time":"12/25/2018
12:34:56","generate_passphrase":true}}'
```

#### Examples

```
# Create - no params
body = {}

# Creating with a peer address and a passphrase
body =
{
  "remote":
    {
        "ip_addresses":["1.2.3.4"]
    }
}

# Creating with a peer name and a generated passphrase that is true
body =
{
  "name":"cp_xyz123",
  "authentication":
    {
        "generate_passphrase":true
    }
}

# Creating with a name, a peer address, and a passphrase
body =
{
  "name":"cp_xyz123",
  "remote":
    {
        "ip_addresses": ["1.2.3.4"]
    },
  "authentication":
    {
        "passphrase":"xyz12345"
    }
 }

# Creating with a proposed encryption protocol
body =
{
  "encryption":
    {
        "proposed":"tls-psk"
    }
}
```

## Create local intercluster LIFs

The local cluster must have an intercluster LIF on each node for the correct operation of cluster peering. If no local intercluster LIFs exist, you can optionally specify LIFs to be created for each node in the local cluster. These local interfaces, if specified, are created on each node before proceeding with the creation of the cluster peering relationship. Cluster peering relationships are not established if there is an error preventing the LIFs from being created. After local interfaces have been created, do not specify them for subsequent cluster peering relationships.

### Local LIF creation fields

- local_network.ip_addresses - List of IP addresses to assign, one per node in the local cluster.
- local_network.netmask - IPv4 mask or subnet mask length.
- local_network.broadcast_domain - Broadcast domain that is in use within the IPspace.
- local_network.gateway - The IPv4 or IPv6 address of the default router.

### Additional information on network routes

When creating LIFs, the network route discovery mechanism might take additional time (1-5 seconds) to become visible in the network outside of the cluster. This delay in publishing the routes might cause an initial cluster peer "create" request to fail. This error disappears with a retry of the same request.

### Example

```
curl -X POST "https://<mgmt-ip>/api/cluster/peers" -d body
```

Note that "<mgmt-ip>" is replaced by the IP address of the cluster management LIF, and "body" is replaced by the JSON body of the POST request containing the fields for the new peering relationship and local LIFs.</mgmt-ip>

### Example POST body

This example shows the POST body when creating four intercluster LIFs on a 4-node cluster before creating a cluster peer relationship.

```
{
  "local_network":
  {
      "interfaces": [
          {"ip_address":"1.2.3.4"},
          {"ip_address":"1.2.3.5"},
          {"ip_address":"1.2.3.6"}
          ],
      "netmask": "255.255.0.0",
      "broadcast_domain": "Default",
      "gateway": "1.2.0.1"
  }
  "remote.ip_addresses": ["1.2.9.9"],
  "authentication.passphrase": "xyz12345"
}
```

## Retrieve a cluster peer

You can retrieve peers in a cluster by issuing a GET request to /cluster/peers. It is also possible to retrieve a specific peer when qualified by its UUID to /cluster/peers/{uuid}. A GET request might have no query parameters or a valid cluster UUID. The former retrieves all records while the latter retrieves the record for the cluster peer with that UUID. The following fields are used for retrieving a cluster peer.

### Required fields

There are no required fields for GET requests.

### Optional fields

The following fields are optional for GET requests

- UUID - UUID of the cluster peer.

### Examples

```
curl -X GET "https://<mgmt-ip>/api/cluster/peers/"
curl -X GET "https://<mgmt-ip>/api/cluster/peers/{uuid}"
curl -X GET "https://<mgmt-ip>/api/cluster/peers/{uuid}?fields=*"
```

## Update a cluster peer

You can update a cluster peer relationship by issuing a PATCH request to /cluster/peers/{uuid}. As in the CLI mode, you can toggle the proposed encryption protocol, update the passphrase, or specify a new set of stable addresses. All PATCH requests take the parameters that are to be updated in the request body. If the

generate_passphrase is "true", the passphrase is returned in the PATCH response. This following fields highlight the parameters that control the modification of an existing cluster peering relationship.

### Required fields

A PATCH request with an empty body has no effect on the cluster peer instance. All other fields and the combinations in which they are valid are indicated below:

- `encryption_proposed` - Toggle the proposed encryption protocol (from "none" to "tls-psk" or otherwise). Authentication must be "true" and a passphrase must be present in body.
- `passphrase`
- `passphrase` or `generate passphrase`
- `remote.ip_addresses`

### Optional fields

- `expiration time` - Set the expiration time of the passphrase.

### Examples

```
# Updating with an empty body
body = {}

# Updating the proposed encryption protocol from tls-psk to none
body =
{
  "authentication":
    {
        "passphrase":"xyz12345",
        "in_use":"ok"
    },
  "encryption":
    {
        "proposed":"none"
    }
}

# Updating the passphrase
body =
{
  "authentication":
   {
       "passphrase":"xyz12345",
       "in_use":"ok"
   }
}

# Setting an auto-generated passphrase
body =
{
  "authentication":
   {
       "generate_passphrase": true,
       "in_use":"ok"
   }
}

# Updating remote IP addresses
body =
{
  "remote":
    {
        "ip_addresses":["10.224.65.30"]
    }
}
```

#### Sample requests

```
# Setting a passphrase
curl -X PATCH 'https://<mgmt-ip>/api/cluster/peers/73123071-d0b9-11e8-
a686-005056a7179a' -d
'{"authentication":{"passphrase":"xyz12345","in_use":"ok"}}'

# Updating a peer address
curl -X PATCH 'https://<mgmt-ip>/api/cluster/peers/73123071-d0b9-11e8-
a686-005056a7179a' -d '{"remote":{"ip_addresses":["1.2.3.4"]}}'
```

---

### Delete a cluster peer

You can delete a cluster peer using the HTTP DELETE request.

#### Required fields

Perform all delete operations on a valid peer UUID. Deleting an invalid peer returns "HTTP 404", which indicates an error.

#### Optional fields

The DELETE operation has no optional fields.

#### Request format

DELETE "https://<mgmt-ip>/api/cluster/peers/{uuid}"</mgmt-ip>

#### Example

The following request deletes a peer with peer UUID "8becc0d4-c12c-11e8-9ceb-005056bbd143".

```
curl -X DELETE "https://<mgmt-ip>/api/cluster/peers/8becc0d4-c12c-11e8-
9ceb-005056bbd143"
```

## Retrieve cluster peers

GET `/cluster/peers`

Retrieves the collection of cluster peers.

### Learn more

- DOC /cluster/peers

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| fields | array[string] | query | False | Specify the fields to return. |
| max_records | integer | query | False | Limit the number of records returned. |
| return_records | boolean | query | False | The default is true for GET calls. When set to false, only the number of records is returned. |
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached. |
| order_by | array[string] | query | False | Order results by specified fields and optional [asc |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| num_records | integer | Number of records |
| records | array[cluster_peer] | |

**Example response**


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "records": {
    "_links": {
      "interfaces": {
        "href": "/api/resourcelink"
      },
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "authentication": {
      "expiry_time": "P1DT2H3M4S or '2017-01-25T11:20:13Z'",
      "in_use": "ok",
      "state": "ok"
    },
    "encryption": {
      "proposed": "none",
      "state": "none"
    },
    "initial_allowed_svms": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "ipspace": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "exchange",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
```

```
      },
      "local_network": {
        "broadcast_domain": "bd1",
        "gateway": "10.1.1.1",
        "interfaces": {
          "ip_address": "10.10.10.7"
        },
        "netmask": "255.255.0.0"
      },
      "name": "cluster2",
      "peer_applications": [
        "snapmirror",
        "flexcache"
      ],
      "remote": {
        "ip_addresses": {
        },
        "name": "cluster2",
        "serial_number": "4048820-60-9"
      },
      "status": {
        "state": "available",
        "update_time": "2017-01-25 11:20:13 UTC"
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
      "version": {
        "full": "NetApp Release 9.4.0: Sun Nov 05 18:20:57 UTC 2017",
        "generation": 9,
        "major": 4,
        "minor": 0
      }
    }
  }
```


## Error

```
Status: Default, Error
```

| Name | Type | Description |
|---|---|---|
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| next | href | |
| self | href | |

_links

| Name | Type | Description |
|---|---|---|
| interfaces | href | |
| self | href | |

authentication

| Name | Type | Description |
|---|---|---|
| expiry_time | string | The time when the passphrase will expire, in ISO 8601 duration format or date and time format. The default is 1 hour. |
| generate_passphrase | boolean | Auto generate a passphrase when true. |
| in_use | string | |
| passphrase | string | A password to authenticate the cluster peer relationship. |
| state | string | |

encryption

| Name | Type | Description |
|---|---|---|
| proposed | string | |
| state | string | |

_links

| Name | Type | Description |
| --- | --- | --- |
| self | href | |

initial_allowed_svms

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

ipspace

The IPspace of the local intercluster LIFs.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | IPspace name |
| uuid | string | IPspace UUID |

interfaces

| Name | Type | Description |
| --- | --- | --- |
| ip_address | string | IPv4 or IPv6 address |

local_network

Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node.

| Name | Type | Description |
| --- | --- | --- |
| broadcast_domain | string | Broadcast domain that is in use within the IPspace. |
| gateway | string | The IPv4 or IPv6 address of the default router. |
| interfaces | array[interfaces] | |
| netmask | string | IPv4 mask or netmask length. |

remote

| Name | Type | Description |
| --- | --- | --- |
| ip_addresses | array[string] | The IPv4 addresses, IPv6 addresses, or hostnames of the peers. |
| name | string | The name of the remote cluster. |
| serial_number | string | The serial number of the remote cluster. |

status

| Name | Type | Description |
| --- | --- | --- |
| state | string | |
| update_time | string | The last time the state was updated. |

version

This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes.

| Name | Type | Description |
| --- | --- | --- |
| full | string | The full cluster version string. |
| generation | integer | The generation portion of the version. |
| major | integer | The major portion of the version. |
| minor | integer | The minor portion of the version. |

cluster_peer

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| authentication | authentication | |
| encryption | encryption | |
| initial_allowed_svms | array[initial_allowed_svms] | The local SVMs allowed to peer with the peer cluster's SVMs. This list can be modified until the remote cluster accepts this cluster peering relationship. |

| Name | Type | Description |
| --- | --- | --- |
| ipspace | ipspace | The IPspace of the local intercluster LIFs. |
| local_network | local_network | Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node. |
| name | string | Optional name for the cluster peer relationship. By default, it is the name of the remote cluster. |
| peer_applications | array[string] | Peering applications against which allowed SVMs are configured. |
| remote | remote | |
| status | status | |
| uuid | string | UUID of the cluster peer relationship. For anonymous cluster peer offers, the UUID will change when the remote cluster accepts the relationship. |
| version | version | This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes. |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
| --- | --- | --- |
| arguments | array[error_arguments] | Message arguments |

| Name | Type | Description |
|---|---|---|
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Create a peering relationship

POST `/cluster/peers`

Creates a peering relationship and, optionally, the IP interfaces it will use. There are two methods used to create a peering relationship:

- Provide a remote IP address - Used when creating a new cluster peer relationship with a specific remote cluster. This requires at least one remote intercluster IP address from the remote cluster.
- Do not provide a remote IP address - Used when the remote IP address is not provided and when the storage system is ready to accept peering requests from foreign clusters.

## Required properties

- `remote.ip_addresses` - Addresses of the remote peers. The local peer must be able to reach and connect to these addresses for the request to succeed in creating a peer. Only required when creating a peering relationship by providing a remote IP address.
- Either set `generate_passphrase` to "true" or provide a passphrase in the body of the request. Only one of these options is required.

## Recommended optional properties

- `name` - Name of the peering relationship or name of the remote peer.
- `passphrase` - User generated passphrase for use in authentication.
- `generate_passphrase` (true/false) - When "true", ONTAP automatically generates a passphrase to authenticate cluster peers.
- `ipspace` - IPspace of the local intercluster LIFs. Assumes Default IPspace if not provided.
- `initial_allowed_svms` - Local SVMs allowed to peer with the peer cluster's SVMs. Can be modified until the remote cluster accepts this cluster peering relationship.
- `local_network` - Fields to create a local intercluster LIF.
- `expiry_time` - Duration in ISO 8601 format for which the user-supplied or auto-generated passphrase is valid. Expiration time must not be greater than seven days into the future. ISO 8601 duration format is "PnDTnHnMnS" or "PnW" where n is a positive integer. The "nD", "nH", "nM" and "nS" fields can be dropped if zero. "P" must always be present and "T" must be present if there are any hours, minutes, or seconds fields.
- `encryption_proposed` (none/tls-psk) - Encryption mechanism of the communication channel between

the two peers.

- `peer_applications` - SVM peering applications (SnapMirror, FlexCache or both) for which the SVM peering relationship is set up.

## Additional information

As with creating a cluster peer through the CLI, the combinations of options must be valid in order for the create operation to succeed. The following list shows the combinations that will succeed and those that will fail:

- A passphrase only (fail)
- A peer IP address (fail)
- A passphrase with an expiration time > 7 days into the future (fail)
- A peer IP address and a passphrase (OK)
- generate_passphrase=true (OK)
- Any proposed encryption protocol (OK)
- An IPspace name or UUID (OK)
- A passphrase, peer IP address, and any proposed encryption protocol (OK)
- A non empty list of initial allowed SVM peer names or UUIDs. (OK)

## Learn more

- DOC /cluster/peers

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| return_records | boolean | query | False | The default is false. If set to true, the records are returned. |

## Request Body

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| authentication | authentication | |
| encryption | encryption | |
| initial_allowed_svms | array[initial_allowed_svms] | The local SVMs allowed to peer with the peer cluster's SVMs. This list can be modified until the remote cluster accepts this cluster peering relationship. |

| Name | Type | Description |
| --- | --- | --- |
| ipspace | ipspace | The IPspace of the local intercluster LIFs. |
| local_network | local_network | Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node. |
| name | string | Optional name for the cluster peer relationship. By default, it is the name of the remote cluster. |
| peer_applications | array[string] | Peering applications against which allowed SVMs are configured. |
| remote | remote | |
| status | status | |
| uuid | string | UUID of the cluster peer relationship. For anonymous cluster peer offers, the UUID will change when the remote cluster accepts the relationship. |
| version | version | This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes. |

**Example request**


```
{
  "_links": {
    "interfaces": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "authentication": {
    "expiry_time": "P1DT2H3M4S or '2017-01-25T11:20:13Z'",
    "in_use": "ok",
    "state": "ok"
  },
  "encryption": {
    "proposed": "none",
    "state": "none"
  },
  "initial_allowed_svms": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "ipspace": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "exchange",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "local_network": {
    "broadcast_domain": "bd1",
    "gateway": "10.1.1.1",
    "interfaces": {
      "ip_address": "10.10.10.7"
    },
    "netmask": "255.255.0.0"
  },
```

```
  "name": "cluster2",
  "peer_applications": [
    "snapmirror",
    "flexcache"
  ],
  "remote": {
    "ip_addresses": {
    },
    "name": "cluster2",
    "serial_number": "4048820-60-9"
  },
  "status": {
    "state": "available",
    "update_time": "2017-01-25 11:20:13 UTC"
  },
  "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
  "version": {
    "full": "NetApp Release 9.4.0: Sun Nov 05 18:20:57 UTC 2017",
    "generation": 9,
    "major": 4,
    "minor": 0
  }
}
```


## Response

```
Status: 201, Created
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| authentication | authentication | |
| ip_address | string | IPv4 or IPv6 address |
| name | string | Optional name for the cluster peer relationship. By default, it is the name of the remote cluster, or a temporary name might be autogenerated for anonymous cluster peer offers. |

#### Example response

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "authentication": {
    "expiry_time": "2017-01-25 11:20:13 UTC"
  },
  "ip_address": "10.10.10.7",
  "name": "cluster2"
}
```

## Error

```
Status: Default
```

ONTAP Error Response Codes

| Error Code | Description |
|---|---|
| 1966366 | The system SVM of the cluster IPspace hosts cluster LIFs only. |
| 4653365 | IPspaces are unavailable with cluster peering: {ipspace}. |
| 4656069 | Specifying a passphrase without remote IP addresses is not supported. |
| 4656070 | The encryption protocol is meaningful only with authenticated cluster peer relationships. |
| 4656071 | Cannot peer with a cluster bearing the same name as the local cluster. |
| 4656072 | The name must conform to the same rules as a cluster name. |
| 4656074 | Cannot check whether all nodes of this cluster support encryption. |
| 4656075 | Cannot specify encryption: this operation requires an ECV of 9.6.0 or later. |
| 4656077 | Specify either remote IP addresses or generate_passphrase. |

| Error Code | Description |
| --- | --- |
| 4656079 | No cluster nodes were found. Check your cluster configuration. |
| 4656081 | Creating an intercluster LIF requires a list of local IP addresses. |
| 4656085 | Cannot create an intercluster LIF with an empty list of local IP addresses. |
| 4656086 | Creating an intercluster LIF requires a broadcast domain that is in use within the IPspace. |
| 4656087 | The number of local intercluster IP addresses must be less than or equal to the number of available nodes. |
| 4656088 | Found no ports matching the IPspace and the broadcast domain. |
| 4656089 | Found no matching entry for IPspace. |
| 4656090 | The given IPspace differs from the IPspace entry found. |
| 4656091 | Creating an intercluster LIF requires a subnet mask or a subnet mask length. |
| 4656096 | Creating an intercluster LIF requires an IPv4 or IPv6 address of the default router. |

| Name | Type | Description |
| --- | --- | --- |
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| interfaces | href | |
| self | href | |

authentication

| Name | Type | Description |
|---|---|---|
| expiry_time | string | The time when the passphrase will expire, in ISO 8601 duration format or date and time format. The default is 1 hour. |
| generate_passphrase | boolean | Auto generate a passphrase when true. |
| in_use | string | |
| passphrase | string | A password to authenticate the cluster peer relationship. |
| state | string | |

encryption

| Name | Type | Description |
|---|---|---|
| proposed | string | |
| state | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

initial_allowed_svms

| Name | Type | Description |
|---|---|---|
| _links | _links | |

| Name | Type | Description |
|---|---|---|
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

ipspace

The IPspace of the local intercluster LIFs.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | IPspace name |
| uuid | string | IPspace UUID |

interfaces

| Name | Type | Description |
|---|---|---|
| ip_address | string | IPv4 or IPv6 address |

local_network

Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node.

| Name | Type | Description |
|---|---|---|
| broadcast_domain | string | Broadcast domain that is in use within the IPspace. |
| gateway | string | The IPv4 or IPv6 address of the default router. |
| interfaces | array[interfaces] | |
| netmask | string | IPv4 mask or netmask length. |

remote

| Name | Type | Description |
|---|---|---|
| ip_addresses | array[string] | The IPv4 addresses, IPv6 addresses, or hostnames of the peers. |

| Name | Type | Description |
| --- | --- | --- |
| name | string | The name of the remote cluster. |
| serial_number | string | The serial number of the remote cluster. |

status

| Name | Type | Description |
| --- | --- | --- |
| state | string | |
| update_time | string | The last time the state was updated. |

version

This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes.

| Name | Type | Description |
| --- | --- | --- |
| full | string | The full cluster version string. |
| generation | integer | The generation portion of the version. |
| major | integer | The major portion of the version. |
| minor | integer | The minor portion of the version. |

cluster_peer

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| authentication | authentication | |
| encryption | encryption | |
| initial_allowed_svms | array[initial_allowed_svms] | The local SVMs allowed to peer with the peer cluster's SVMs. This list can be modified until the remote cluster accepts this cluster peering relationship. |
| ipspace | ipspace | The IPspace of the local intercluster LIFs. |

| Name | Type | Description |
| --- | --- | --- |
| local_network | local_network | Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node. |
| name | string | Optional name for the cluster peer relationship. By default, it is the name of the remote cluster. |
| peer_applications | array[string] | Peering applications against which allowed SVMs are configured. |
| remote | remote | |
| status | status | |
| uuid | string | UUID of the cluster peer relationship. For anonymous cluster peer offers, the UUID will change when the remote cluster accepts the relationship. |
| version | version | This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes. |

authentication

| Name | Type | Description |
| --- | --- | --- |
| expiry_time | string | The date and time the passphrase will expire. The default expiry time is one hour. |
| passphrase | string | A password to authenticate the cluster peer relationship. |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |

| Name | Type | Description |
|---|---|---|
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Delete a cluster peer

DELETE `/cluster/peers/{uuid}`

Deletes a cluster peer.

## Learn more

- DOC /cluster/peers

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| uuid | string | path | True | |

## Response

```
Status: 200, Ok
```

## Error

```
Status: Default
```

ONTAP Error Response Codes

| Error Code | Description |
| --- | --- |
| 4663070 | Unable to delete cluster peer relationship due to an ongoing Vserver migration. |

| Name | Type | Description |
| --- | --- | --- |
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Retrieve a cluster peer instance

GET `/cluster/peers/{uuid}`

Retrieves a specific cluster peer instance.

## Learn more

- DOC /cluster/peers

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| uuid | string | path | True | Cluster peer relationship UUID |
| fields | array[string] | query | False | Specify the fields to return. |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| authentication | authentication | |
| encryption | encryption | |
| initial_allowed_svms | array[initial_allowed_svms] | The local SVMs allowed to peer with the peer cluster's SVMs. This list can be modified until the remote cluster accepts this cluster peering relationship. |
| ipspace | ipspace | The IPspace of the local intercluster LIFs. |
| local_network | local_network | Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node. |
| name | string | Optional name for the cluster peer relationship. By default, it is the name of the remote cluster. |
| peer_applications | array[string] | Peering applications against which allowed SVMs are configured. |
| remote | remote | |
| status | status | |
| uuid | string | UUID of the cluster peer relationship. For anonymous cluster peer offers, the UUID will change when the remote cluster accepts the relationship. |
| version | version | This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes. |

**Example response**


```
{
  "_links": {
    "interfaces": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "authentication": {
    "expiry_time": "P1DT2H3M4S or '2017-01-25T11:20:13Z'",
    "in_use": "ok",
    "state": "ok"
  },
  "encryption": {
    "proposed": "none",
    "state": "none"
  },
  "initial_allowed_svms": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "ipspace": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "exchange",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "local_network": {
    "broadcast_domain": "bd1",
    "gateway": "10.1.1.1",
    "interfaces": {
      "ip_address": "10.10.10.7"
    },
    "netmask": "255.255.0.0"
  },
```

```
    "name": "cluster2",
    "peer_applications": [
      "snapmirror",
      "flexcache"
    ],
    "remote": {
      "ip_addresses": {
      },
      "name": "cluster2",
      "serial_number": "4048820-60-9"
    },
    "status": {
      "state": "available",
      "update_time": "2017-01-25 11:20:13 UTC"
    },
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
    "version": {
      "full": "NetApp Release 9.4.0: Sun Nov 05 18:20:57 UTC 2017",
      "generation": 9,
      "major": 4,
      "minor": 0
    }
  }
```


## Error

```
Status: Default, Error
```

| Name | Type | Description |
|---|---|---|
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| interfaces | href | |
| self | href | |

authentication

| Name | Type | Description |
|---|---|---|
| expiry_time | string | The time when the passphrase will expire, in ISO 8601 duration format or date and time format. The default is 1 hour. |
| generate_passphrase | boolean | Auto generate a passphrase when true. |
| in_use | string | |
| passphrase | string | A password to authenticate the cluster peer relationship. |
| state | string | |

encryption

| Name | Type | Description |
|---|---|---|
| proposed | string | |
| state | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

initial_allowed_svms

| Name | Type | Description |
|---|---|---|
| _links | _links | |

| Name | Type | Description |
|---|---|---|
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

ipspace

The IPspace of the local intercluster LIFs.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | IPspace name |
| uuid | string | IPspace UUID |

interfaces

| Name | Type | Description |
|---|---|---|
| ip_address | string | IPv4 or IPv6 address |

local_network

Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node.

| Name | Type | Description |
|---|---|---|
| broadcast_domain | string | Broadcast domain that is in use within the IPspace. |
| gateway | string | The IPv4 or IPv6 address of the default router. |
| interfaces | array[interfaces] | |
| netmask | string | IPv4 mask or netmask length. |

remote

| Name | Type | Description |
|---|---|---|
| ip_addresses | array[string] | The IPv4 addresses, IPv6 addresses, or hostnames of the peers. |

| Name | Type | Description |
|---|---|---|
| name | string | The name of the remote cluster. |
| serial_number | string | The serial number of the remote cluster. |

status

| Name | Type | Description |
|---|---|---|
| state | string | |
| update_time | string | The last time the state was updated. |

version

This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes.

| Name | Type | Description |
|---|---|---|
| full | string | The full cluster version string. |
| generation | integer | The generation portion of the version. |
| major | integer | The major portion of the version. |
| minor | integer | The minor portion of the version. |

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |

| Name | Type | Description |
|---|---|---|
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Update a cluster peer instance

PATCH `/cluster/peers/{uuid}`

Updates a cluster peer instance.

## Learn more

- DOC /cluster/peers

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| uuid | string | path | True | Cluster peer relationship UUID |

## Request Body

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| authentication | authentication | |
| encryption | encryption | |
| initial_allowed_svms | array[initial_allowed_svms] | The local SVMs allowed to peer with the peer cluster's SVMs. This list can be modified until the remote cluster accepts this cluster peering relationship. |
| ipspace | ipspace | The IPspace of the local intercluster LIFs. |
| local_network | local_network | Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node. |

| Name | Type | Description |
| --- | --- | --- |
| name | string | Optional name for the cluster peer relationship. By default, it is the name of the remote cluster. |
| peer_applications | array[string] | Peering applications against which allowed SVMs are configured. |
| remote | remote | |
| status | status | |
| uuid | string | UUID of the cluster peer relationship. For anonymous cluster peer offers, the UUID will change when the remote cluster accepts the relationship. |
| version | version | This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes. |

**Example request**


```
{
  "_links": {
    "interfaces": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "authentication": {
    "expiry_time": "P1DT2H3M4S or '2017-01-25T11:20:13Z'",
    "in_use": "ok",
    "state": "ok"
  },
  "encryption": {
    "proposed": "none",
    "state": "none"
  },
  "initial_allowed_svms": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "ipspace": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "exchange",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "local_network": {
    "broadcast_domain": "bd1",
    "gateway": "10.1.1.1",
    "interfaces": {
      "ip_address": "10.10.10.7"
    },
    "netmask": "255.255.0.0"
  },
```

```
  "name": "cluster2",
  "peer_applications": [
    "snapmirror",
    "flexcache"
  ],
  "remote": {
    "ip_addresses": {
    },
    "name": "cluster2",
    "serial_number": "4048820-60-9"
  },
  "status": {
    "state": "available",
    "update_time": "2017-01-25 11:20:13 UTC"
  },
  "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
  "version": {
    "full": "NetApp Release 9.4.0: Sun Nov 05 18:20:57 UTC 2017",
    "generation": 9,
    "major": 4,
    "minor": 0
  }
}
```


## Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| authentication | authentication | |
| ip_address | string | IPv4 or IPv6 address |
| name | string | Optional name for the cluster peer relationship. By default, it is the name of the remote cluster, or a temporary name might be autogenerated for anonymous cluster peer offers. |

#### Example response

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "authentication": {
    "expiry_time": "2017-01-25 11:20:13 UTC"
  },
  "ip_address": "10.10.10.7",
  "name": "cluster2"
}
```

## Error

```
Status: Default
```

ONTAP Error Response Codes

| Error Code | Description |
|---|---|
| 4653261 | Error finding IPspace. |
| 4655058 | Expiration time cannot be more than 7 days in the future. |
| 4656070 | The encryption protocol is meaningful only with authenticated cluster peer relationships. |
| 4656072 | The name must conform to the same rules as a cluster name. |
| 4656073 | Changing the encryption state requires the refreshing of the authentication passphrase. |
| 4656075 | Cannot specify encryption: this operation requires an ECV of ONTAP 9.6.0 or later. |
| 4656076 | Cluster peer modify was attempted with mismatched IPv4 and IPv6 addresses. |
| 4656081 | The remote IP address list is empty. |
| 4656082 | Specify either a passphrase or "-generate-passphrase". |

| Error Code | Description |
| --- | --- |
| 4656083 | Cannot auto-generate a passphrase when "generate-passphrase" is false. Modifying a passphrase using an auto-generated passphrase requires "generate-passphrase" be true. |
| 4656084 | Passphrase can only be modified with an authenticated cluster peer relationship. |
| 4656092 | Cluster peer modify was attempted with a host name that did not resolve to an IPv4 or IPv6 address. |
| 4656095 | The address family of the specified peer addresses is not valid in this IPspace. Use /api/network/interfaces/ to verify that required LIFs are present and operational on each cluster node. |

| Name | Type | Description |
| --- | --- | --- |
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| interfaces | href | |
| self | href | |

authentication

| Name | Type | Description |
|---|---|---|
| expiry_time | string | The time when the passphrase will expire, in ISO 8601 duration format or date and time format. The default is 1 hour. |
| generate_passphrase | boolean | Auto generate a passphrase when true. |
| in_use | string | |
| passphrase | string | A password to authenticate the cluster peer relationship. |
| state | string | |

encryption

| Name | Type | Description |
|---|---|---|
| proposed | string | |
| state | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

initial_allowed_svms

| Name | Type | Description |
|---|---|---|
| _links | _links | |

| Name | Type | Description |
|---|---|---|
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

ipspace

The IPspace of the local intercluster LIFs.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | IPspace name |
| uuid | string | IPspace UUID |

interfaces

| Name | Type | Description |
|---|---|---|
| ip_address | string | IPv4 or IPv6 address |

local_network

Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node.

| Name | Type | Description |
|---|---|---|
| broadcast_domain | string | Broadcast domain that is in use within the IPspace. |
| gateway | string | The IPv4 or IPv6 address of the default router. |
| interfaces | array[interfaces] | |
| netmask | string | IPv4 mask or netmask length. |

remote

| Name | Type | Description |
|---|---|---|
| ip_addresses | array[string] | The IPv4 addresses, IPv6 addresses, or hostnames of the peers. |

| Name | Type | Description |
| --- | --- | --- |
| name | string | The name of the remote cluster. |
| serial_number | string | The serial number of the remote cluster. |

status

| Name | Type | Description |
| --- | --- | --- |
| state | string | |
| update_time | string | The last time the state was updated. |

version

This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes.

| Name | Type | Description |
| --- | --- | --- |
| full | string | The full cluster version string. |
| generation | integer | The generation portion of the version. |
| major | integer | The major portion of the version. |
| minor | integer | The minor portion of the version. |

cluster_peer

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| authentication | authentication | |
| encryption | encryption | |
| initial_allowed_svms | array[initial_allowed_svms] | The local SVMs allowed to peer with the peer cluster's SVMs. This list can be modified until the remote cluster accepts this cluster peering relationship. |
| ipspace | ipspace | The IPspace of the local intercluster LIFs. |

| Name | Type | Description |
| --- | --- | --- |
| local_network | local_network | Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node. |
| name | string | Optional name for the cluster peer relationship. By default, it is the name of the remote cluster. |
| peer_applications | array[string] | Peering applications against which allowed SVMs are configured. |
| remote | remote | |
| status | status | |
| uuid | string | UUID of the cluster peer relationship. For anonymous cluster peer offers, the UUID will change when the remote cluster accepts the relationship. |
| version | version | This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes. |

authentication

| Name | Type | Description |
| --- | --- | --- |
| expiry_time | string | The date and time the passphrase will expire. The default expiry time is one hour. |
| passphrase | string | A password to authenticate the cluster peer relationship. |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |

| Name | Type | Description |
| --- | --- | --- |
| message | string | Message argument |

error

| Name | Type | Description |
| --- | --- | --- |
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

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